FMEA

PREVENT RISK

COURSE SLIDES · WSQ

Continuous Process Improvement with FMEA

WSQ Course Code: TGS-2021009031

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v11.0 · 13 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

AM attendance

Scan the SSG digital-attendance QR code when class starts.

2

PM attendance

Scan again after lunch when instructed.

3

Assessment attendance

Complete the separate attendance before assessment.

4

Funding requirement

At least 75% attendance and a Competent result are required.

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About the Trainer



Trainer Profile Template

Trainer completes this profile before delivery

NAME
[Trainer name]

DESIGNATION
[Title / role]

QUALIFICATIONS
[Relevant qualifications]

EXPERTISE
[Subject-matter strengths]

EXPERIENCE
[Industry and training experience]

CONTACT
[Approved contact channel]

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer and courseware developer

EXPERTISE
Process improvement, analytics, automation and AI

DELIVERS
WSQ data, software, cloud and AI programmes

FOUNDER
Tertiary Infotech Academy / Tertiary Courses

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INTRODUCTION

Let's Know Each Other

You

- Name and organisation
- Role and process context
- One improvement challenge

Your process

- Customer and output
- One recurring failure
- Current measure

Your goal

- Risk to prevent
- Capability to improve
- Learning outcome for these two days

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HOUSEKEEPING

Ground Rules

1. Set mobile phones to silent mode.
2. Participate actively—no question is too small.
3. Respect different views; agree to disagree.
4. One conversation at a time.
5. Return from breaks on time.
6. Maintain at least 75% attendance.

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FIVE CONNECTED TOPICS · ONE ORION CASE

Course Outline

1

Topic 1
Introduction to Continuous Process Improvement

2

Topic 2
Root Cause Analysis

3

Topic 3
Design and Apply FMEA

4

Topic 4
Identify and Direct Improvement Opportunities

5

Topic 5
Manage Improvement with a Process Control Plan

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16 HOURS · 13.5 HOURS TRAINING + 2.5 HOURS ASSESSMENT

Two-Day Learning Journey

1

Day 1 · AM

Administration, Topic 1 and Activities 1–2: CI system, SIPOC and PDCA.

2

Day 1 · PM

Topic 2 and Activities 3–4: Pareto, Fishbone, 5 Whys and Fault Tree.

3

Day 2 · AM

Topic 3 and Activities 5–6: build, rate and optimise the Orion PFMEA.

4

Day 2 · PM

Topics 4–5, Activities 7–10, then 60-minute WA and 90-minute PP.

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BY THE END OF THE COURSE

Learning Outcomes

1

ELO1

Create a continuous process improvement system.

2

ELO2

Perform root cause analysis for continuous process improvement.

3

ELO3

Design and apply Failure Mode and Effects Analysis (FMEA).

4

ELO4

Identify and direct improvement opportunities.

5

ELO5

Manage and evaluate improvement with a Process Control Plan.

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Skills Framework Alignment

Knowledge K1–K4

- K1: Identification of continuous improvement opportunities
- K2: Direct continuous improvement activities
- K3: Management of systems and processes to facilitate continuous improvement
- K4: Monitoring and evaluation of continuous improvement processes

Abilities A1–A10

- A1: Manage and implement continuous improvement systems
- A2: Promote team support and coach team members toward continuous improvement
- A3: Gather, access and record production information to track continuous improvements
- A4: Analyse, synthesise and interpret information
- A5: Design and apply improvement tools and strategies
- A6: Monitor and evaluate improvement processes
- A7: Identify continuous improvement opportunities
- A8: Discuss and direct continuous improvement activities
- A9: Manage systems and processes to facilitate continued improvement
- A10: Evaluate improvement processes

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BEFORE ASSESSMENT

Briefing for Assessment

1

Individual work

No discussion or copying during assessment.

2

Open book

Use course slides, Learner Guide and approved materials.

3

Evidence

Complete every response and practical deliverable.

4

Conduct

No photos; phones and unrelated materials remain away.

5

Writing

Use black/blue pen where hard-copy responses are required.

6

Rights

Re-assessment and appeal processes are available.

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WA + PP · 150 MINUTES

Final Assessment

Written Assessment

- 4 open-ended questions
- K1–K4
- 60 minutes

Practical Performance

- 5 connected tasks
- A1–A10
- 90 minutes

Competency

- Open book
- C / NYC outcome
- Submit evidence and sign record

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INDIVIDUAL · OPEN BOOK · COMPETENT / NOT YET COMPETENT

Assessment Flow

1. Assessment attendance
2. Written Assessment · K1–K4 · 60 min
3. Practical Performance · A1–A10 · 90 min
4. Submit candidate evidence
5. Sign Assessment Summary Record

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COURSE REPOSITORY

Access the Hands-On Activities

Download

- Open the GitHub repository
- Choose Code → Download ZIP
- Extract before editing files

Work

- One folder per activity
- Use the starter workbook
- Follow detailed steps in the Learner Guide

Verify

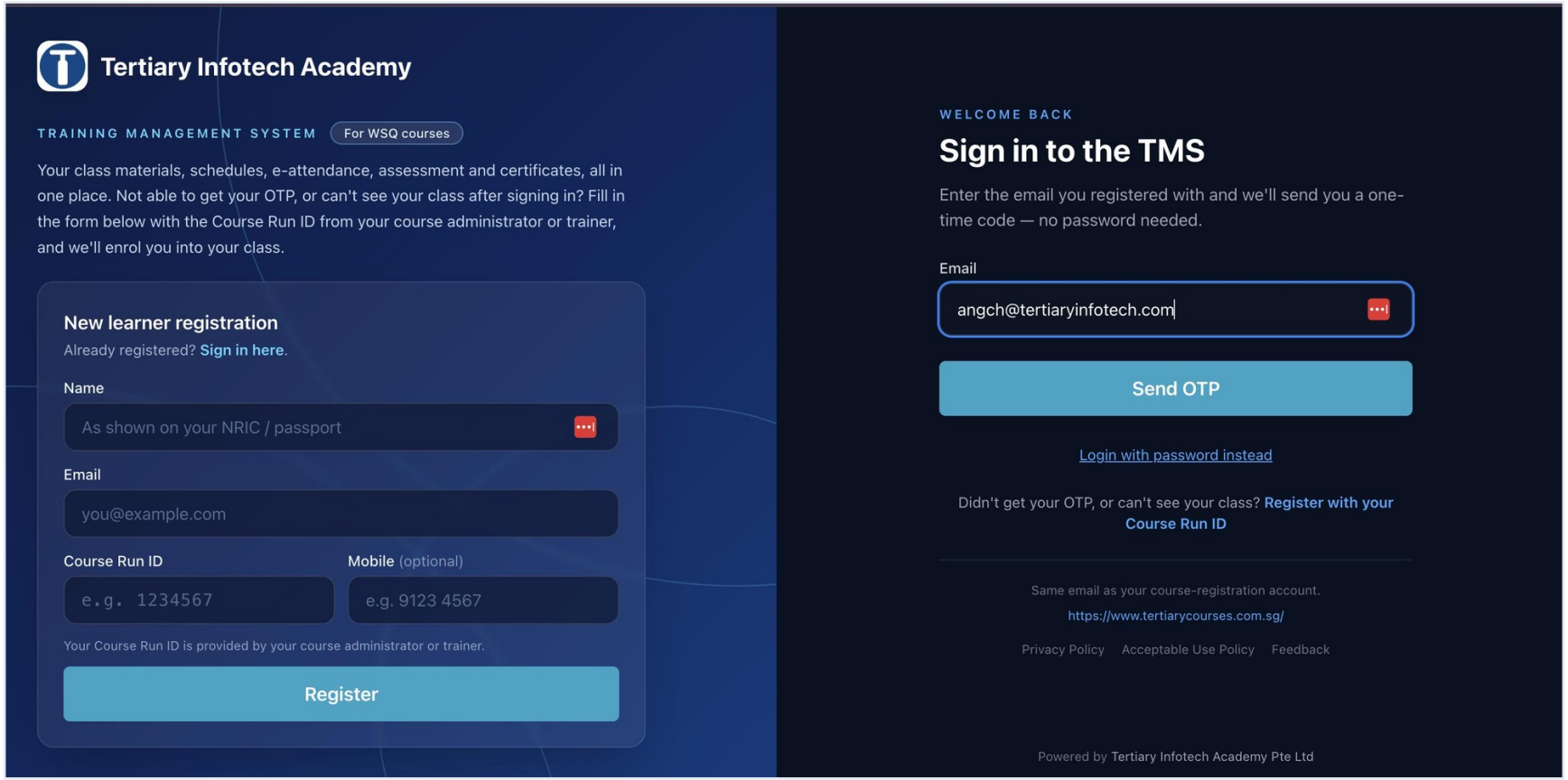
- Run the folder checklist
- Compare with the completed sample
- Save your evidence clearly

https://github.com/tertiarycourses/TGS-2021009031-Continuous-Process-Improvement-with-FMEA

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COURSE PORTAL

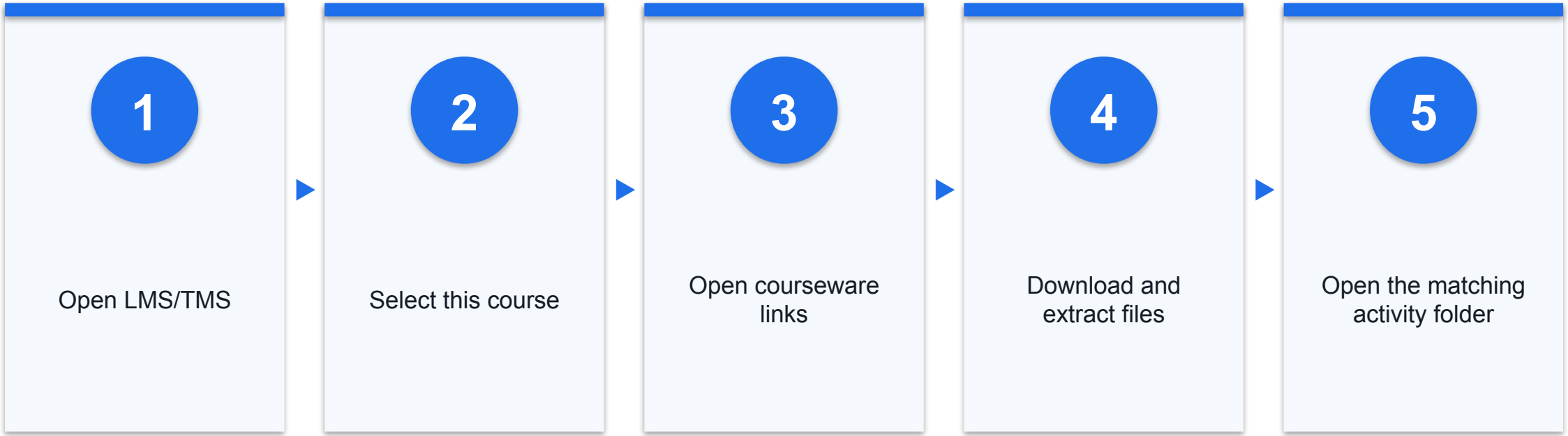
Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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Download Course Materials



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PRACTICE EXAM PORTAL

Practice and Readiness Check

Review

- Revisit K1–K4 and A1–A10
- Open completed activity evidence
- Explain decisions, not only scores

Practise

- Use the assigned FMEA practice set
- Work individually first
- Retain formulas and rationale

Reflect

- Trace action to failure mechanism
- Check residual risk and controls
- State one limitation and next step

https://exams.tertiaryinfotech.com/

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ONE CASE · TEN ACTIVITIES

Connected Case — Orion Precision Electronics

1

Process

Surface-mount technology (SMT) assembly of industrial sensor-controller boards

2

Signal

Rising solder defects, rework, late lots and two customer returns after a new product ramp-up

3

Goal

Reduce first-pass defects, stabilise cycle time and prevent recurrence through a living PFMEA and Control Plan

4

Learning path

Define → analyse → anticipate → improve → control

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01

TOPIC 01

Introduction to Continuous Process Improvement

Culture · governance · PDCA · process definition

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ELO1 · A1 · A2 · K1

Key Concepts — Continuous Improvement

1

System, not event

Continuous improvement is a managed cycle of evidence, action, learning and standardisation.

2

People closest to work

Operators and technicians surface weak signals and practical constraints.

3

PDCA discipline

Test a defined change, compare evidence and standardise only what works.

4

Shared process view

SIPOC and a process map establish boundaries before analysis begins.

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TOPIC 01 · WHY IT MATTERS

Improvement is a management system

A workshop creates ideas; a system creates repeatable learning, ownership and control.

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TOPIC 01

What continuous improvement changes

1

Performance

Reduce defects, delay, cost and instability against a defined baseline.

2

Capability

Build reliable processes that repeatedly meet customer and regulatory needs.

3

Learning

Turn experiments, failures and operator knowledge into standard work.

4

Culture

Make problem visibility safe and improvement part of normal work.

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TOPIC 01

Continuous improvement and breakthrough change

Use

- Frequent, evidence-led adjustments
- Lower implementation risk
- Builds habits and process knowledge

Contrast / check

- Larger redesign or technology shift
- Higher uncertainty and coordination
- May be necessary when the current system is structurally incapable

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TOPIC 01

Lean, Six Sigma and FMEA

Lean

- Improve flow
- Remove non-value work
- Shorten lead time

Six Sigma

- Reduce variation
- Use data to test causes
- Improve capability

FMEA

- Anticipate failure
- Prioritise prevention
- Link risk to controls

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TOPIC 01

A culture that can improve

1

Will

Leaders make priorities and trade-offs explicit.

2

Skill

Teams know how to frame, analyse, test and control.

3

Authority

Owners can act within agreed boundaries.

4

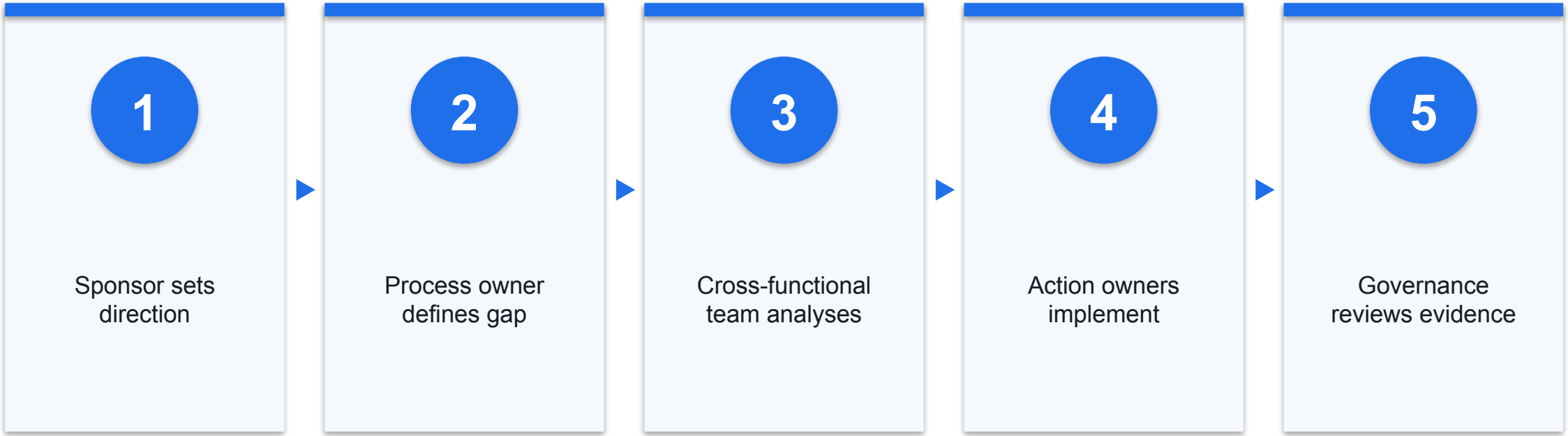
Guidance

Coaching and standards turn tools into good judgement.

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TOPIC 01

Continuous improvement governance



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TOPIC 01

PDCA turns change into learning

1

Plan

define gap, target and hypothesis

2

Do

test at controlled scale

LEARN ↻

4

Act

standardise, adjust or abandon

3

Check

compare evidence and side effects

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TOPIC 01

Plan with evidence

1

Problem

Current versus required performance, with scope and time.

2

Baseline

Trusted operational data and agreed definitions.

3

Target

Specific outcome, measure and time horizon.

4

Hypothesis

Why the proposed countermeasure should change the result.

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TOPIC 01

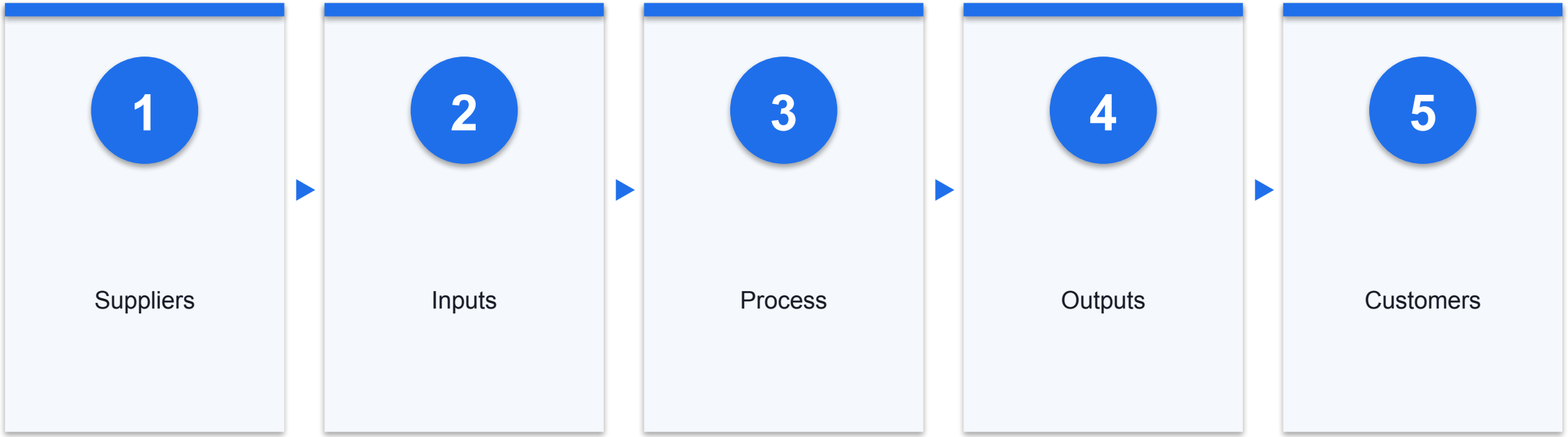
Team roles in improvement

1. **Sponsor**
   Removes barriers and protects the business priority.
2. **Process owner**
   Owns performance and the post-project control system.
3. **Facilitator**
   Keeps the method rigorous and participation balanced.
4. **Operators and experts**
   Provide actual-work knowledge and test feasibility.
5. **Data owner**
   Protects definitions, lineage and evidence quality.
6. **Action owner**
   Implements, verifies and documents a countermeasure.

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TOPIC 01

SIPOC establishes the analysis boundary



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TOPIC 01

Voice of customer and voice of process

Use

- Needs, complaints and field effects
- Critical-to-quality expectations
- What failure means to the user

Contrast / check

- Yield, cycle time and defect mechanisms
- Variation, controls and capability
- What the operation can repeatedly deliver

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TOPIC 01

Coaching for team support

1

Ask before telling

Use questions to surface assumptions and evidence.

2

Make work visible

Use boards, owners and dates so risk is shared.

3

Reinforce learning

Review what changed and why, not only the result.

4

Protect candour

Treat an exposed weakness as useful information.

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ACTIVITY 01 · A1, A2, K1

Launch the Orion Continuous Improvement System

ORION SCENARIO

Orion's SMT line has competing priorities, informal escalation and no shared definition of success.

YOUR MISSION

Build a one-page improvement charter and team operating system for the connected Orion case.

DELIVERABLE

CI charter, role map, coaching cadence and daily/weekly review board

TOOLS AND FILES

Excel charter workbook · CI coaching checklist · PDCA governance

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Activity 01 — Decision Target

1

Objective

Create governance, coaching and visible management for the improvement effort.

2

Question 1

Which decision belongs to the sponsor rather than the team?

3

Question 2

How will the process owner know the improvement is sustained?

4

Evidence standard

A reviewer can identify the gap, target, measures, owners, cadence and coaching approach without asking for missing definitions.

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REVIEW BEFORE SUBMISSION

Activity 01 — Learning Check

✓ Acceptance check

A reviewer can identify the gap, target, measures, owners, cadence and coaching approach without asking for missing definitions.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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ACTIVITY 02 · A1, A2, K1

Define the Process with SIPOC and PDCA

ORION SCENARIO

The team disagrees about whether stencil preparation, printing, placement or reflow belongs in scope.

YOUR MISSION

Build an agreed SIPOC, high-level flow and PDCA experiment plan before root-cause analysis.

DELIVERABLE

SIPOC, five-stage SMT process map, stakeholder needs and PDCA experiment canvas

TOOLS AND FILES

SIPOC web tool · Excel workbook · PDCA hypothesis

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Activity 02 — Decision Target

1

Objective

Set a process boundary and convert the improvement goal into a testable PDCA plan.

2

Question 1

What is deliberately out of scope?

3

Question 2

Which requirement will later appear as a function in the PFMEA?

4

Evidence standard

The SIPOC boundary is unambiguous and the PDCA hypothesis contains change, predicted outcome, measure and guardrail.

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REVIEW BEFORE SUBMISSION

Activity 02 — Learning Check

✓ Acceptance check

The SIPOC boundary is unambiguous and the PDCA hypothesis contains change, predicted outcome, measure and guardrail.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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TOPIC RECAP

Recap — Introduction to Continuous Process Improvement

1

Concept

Continuous improvement is a managed cycle of evidence, action, learning and standardisation.

2

Decision

Operators and technicians surface weak signals and practical constraints.

3

Application

CI charter, role map, coaching cadence and daily/weekly review board

4

Transfer

Name one workplace risk, evidence source and next action.

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02

TOPIC 02

Root Cause Analysis

Evidence · Pareto · Fishbone · 5 Whys · Fault Tree

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ELO2 · A3 · A4

Key Concepts — Root Cause Analysis

1

Describe the gap

A good problem statement names the measure, location, timing and magnitude without guessing the cause.

2

Separate cause from symptom

The customer effect, process failure and causal mechanism belong at different levels.

3

Triangulate

Use data prioritisation, structured brainstorming and verification together.

4

Mechanism over blame

Useful causes describe how the process creates failure and point toward controllable action.

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TOPIC 02 · WHY IT MATTERS

A root cause is a verified mechanism

It is not the first plausible explanation, a correlation, or the person nearest the failure.

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TOPIC 02

Build a neutral problem statement

1

What

The measurable defect, delay or performance gap.

2

Where

Process step, line, product family or customer segment.

3

When

Start, pattern, shift, period or change point.

4

How much

Magnitude, rate, trend and business/customer impact.

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TOPIC 02

Cause, failure and consequence

Use

- Cause: stencil aperture partly blocked
- Mechanism: insufficient paste transfer
- Controllable through cleaning and verification

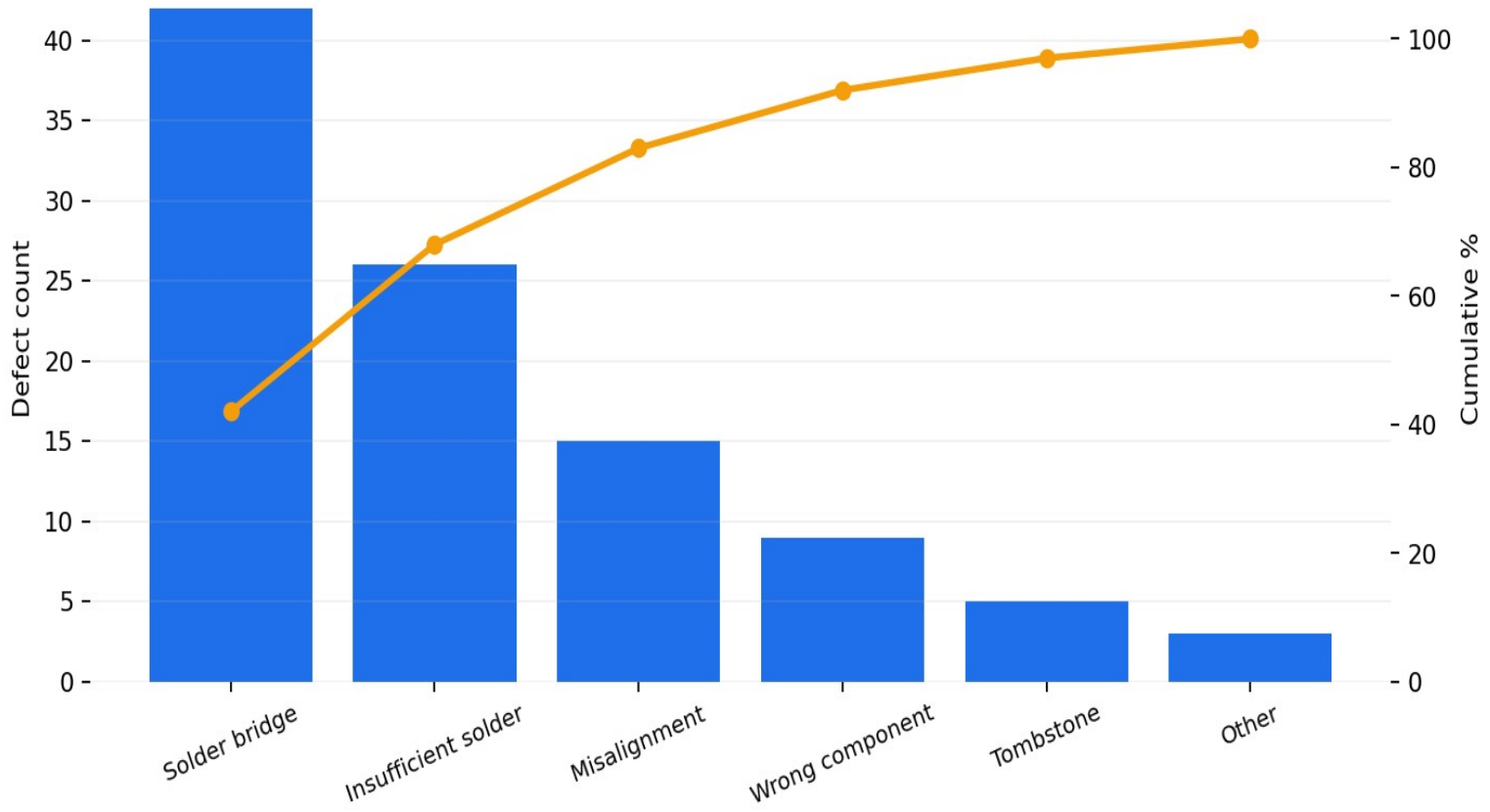
Contrast / check

- Failure mode: insufficient solder joint
- Effect: intermittent signal or field return
- Observed symptom is not automatically the root cause

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TOPIC 02

Pareto focuses the investigation



READ THE SIGNAL

- Solder bridge is the largest category.
- The cumulative line shows concentration.
- Priority is not proof of cause.

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TOPIC 02

Read a Pareto chart correctly

1

Rank

Bars order categories by frequency or impact.

2

Cumulative line

Shows how quickly the categories account for the total.

3

Stratify

Check shift, product and machine before combining data.

4

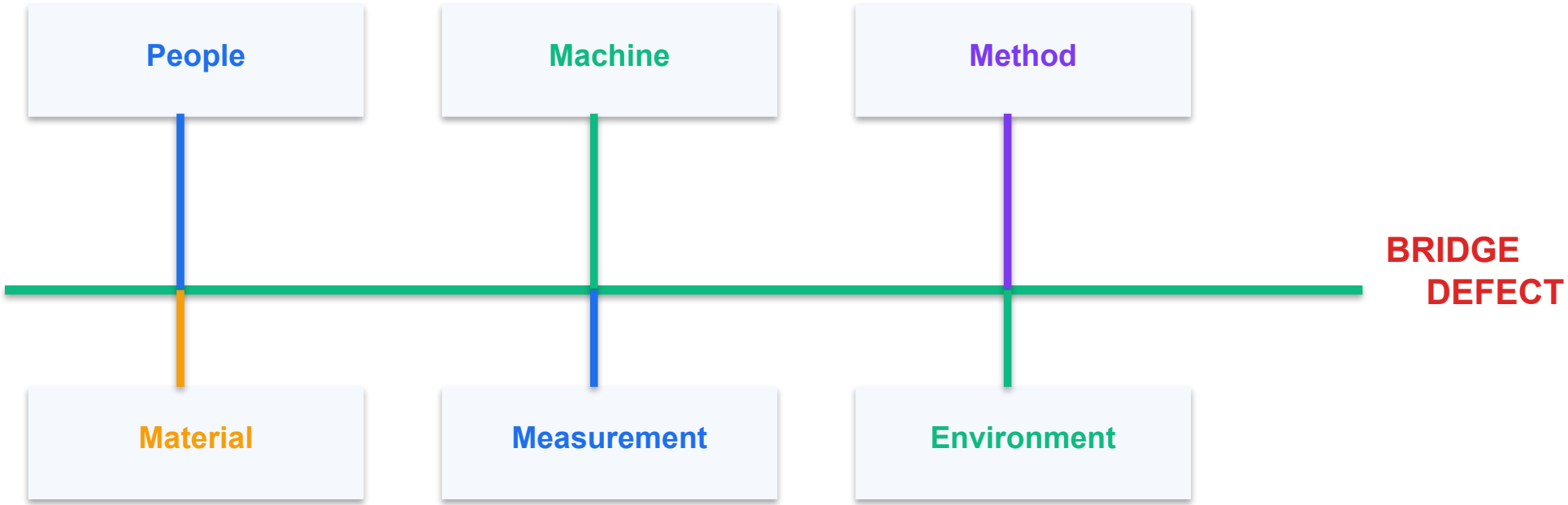
Do not overclaim

The largest category is a priority, not yet a cause.

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TOPIC 02

Fishbone structures possible causes



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TOPIC 02

From brainstorm to evidence

1

Candidate cause

Write a specific mechanism, not a vague label.

2

Prediction

State what data pattern would be expected if it were true.

3

Test

Compare lots, shifts, machines, materials or controlled trials.

4

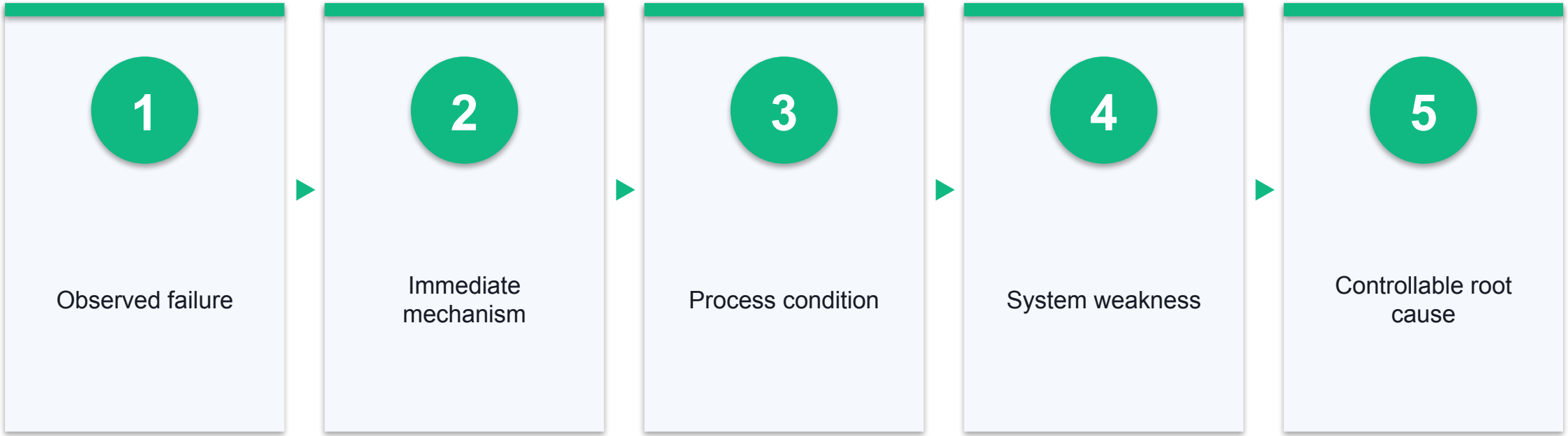
Decision

Confirm, reject or keep open with the next evidence needed.

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TOPIC 02

Five Whys follows one causal chain



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TOPIC 02

Strong and weak 5 Whys

Use

- Each answer is evidence-supported
- The chain stays on one mechanism
- The endpoint suggests a system change

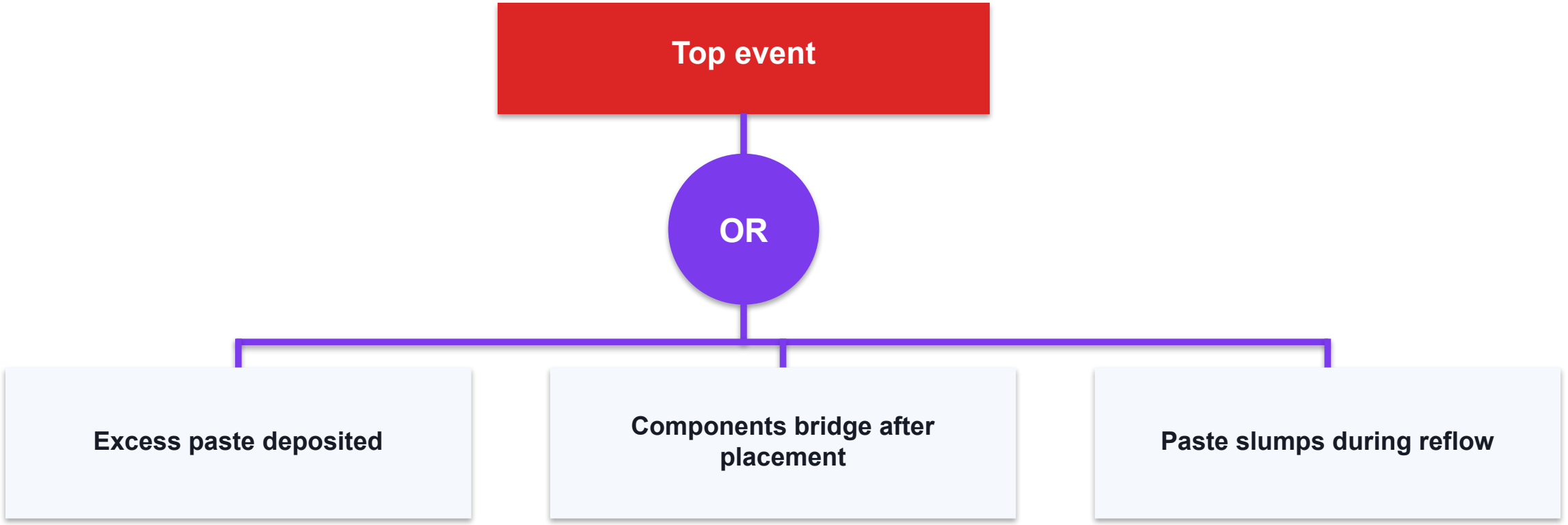
Contrast / check

- Jumps between unrelated branches
- Ends with 'operator error'
- Treats five as a magic number

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TOPIC 02

Fault Tree Analysis tests combinations



Gates make causal combinations explicit; evidence tests the basic events.

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TOPIC 02

Fishbone, 5 Whys and Fault Tree

Use

- Fishbone widens the search
- 5 Whys deepens one likely chain
- Use when mechanisms are still being explored

Contrast / check

- Fault Tree formalises logical combinations
- Supports safety and reliability reasoning
- Use when several paths can create the top event

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TOPIC 02

Data quality checks before conclusion

1

Definition

Are defect categories and denominators consistent?

2

Measurement

Can inspectors or sensors reproduce the classification?

3

Completeness

Are rework, scrap and escaped defects all represented?

4

Context

Are product mix, shift and process changes recorded?

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ACTIVITY 03 · A3, A4

Prioritise the Defect Pattern with Pareto Analysis

ORION SCENARIO

Orion has 240 inspection records across products, shifts and printing machines but anecdotes point in different directions.

YOUR MISSION

Validate the dataset, build a Pareto view and identify the defect/stratum requiring deeper analysis.

DELIVERABLE

Clean defect log, frequency summary, cumulative percentage and interactive Pareto chart

TOOLS AND FILES

Excel Table · COUNTIFS · SUM · cumulative percentage · Pareto chart

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Activity 03 — Decision Target

1

Objective

Gather, validate and interpret production information to focus the investigation.

2

Question 1

What is the leading defect and how much of the total does it represent?

3

Question 2

Which stratum shows the strongest signal, and what limitation remains?

4

Evidence standard

The Pareto counts reconcile to the source, cumulative percentage reaches 100%, and the conclusion distinguishes priority from cause.

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REVIEW BEFORE SUBMISSION

Activity 03 — Learning Check

✓ Acceptance check

The Pareto counts reconcile to the source, cumulative percentage reaches 100%, and the conclusion distinguishes priority from cause.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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ACTIVITY 04 · A3, A4

Verify Root Causes with Fishbone, 5 Whys and Fault Tree

ORION SCENARIO

Solder bridging is concentrated on Printer P2 during the late shift, but the causal mechanism is not yet verified.

YOUR MISSION

Use three complementary RCA tools, then convert candidate causes into evidence tests and corrective-action hypotheses.

DELIVERABLE

6M Fishbone, evidence-backed 5 Whys, fault tree and cause-verification matrix

TOOLS AND FILES

Fishbone tool · 5 Whys tool · Fault Tree · verification planning

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Activity 04 — Decision Target

1

Objective

Analyse evidence and build a defensible causal explanation for the leading defect.

2

Question 1

Which cause is a mechanism rather than a label?

3

Question 2

What observation would disconfirm the preferred explanation?

4

Evidence standard

At least one cause is supported by a reproducible evidence test, the fault-tree logic is valid, and no conclusion relies on blame.

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REVIEW BEFORE SUBMISSION

Activity 04 — Learning Check

✓ Acceptance check

At least one cause is supported by a reproducible evidence test, the fault-tree logic is valid, and no conclusion relies on blame.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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TOPIC RECAP

Recap — Root Cause Analysis

1

Concept

A good problem statement names the measure, location, timing and magnitude without guessing the cause.

2

Decision

The customer effect, process failure and causal mechanism belong at different levels.

3

Application

Clean defect log, frequency summary, cumulative percentage and interactive Pareto chart

4

Transfer

Name one workplace risk, evidence source and next action.

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Day 2

FMEA design · improvement action · process control

TOPIC 03

Design and Apply FMEA

Scope · function · failure chain · risk · action · learning

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ELO3 · A5 · A6 · K3

Key Concepts — Design and Apply FMEA

1

Prospective thinking

FMEA asks what could fail before the next customer or process loss occurs.

2

Failure chain

Function → failure mode → effect → cause → controls makes risk traceable.

3

Risk-informed action

Severity, occurrence and detection guide judgement; RPN alone never grants permission to ignore risk.

4

Living knowledge

Actions, verification, re-rating and review triggers keep the FMEA current.

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TOPIC 03 · WHY IT MATTERS

FMEA is prevention made visible

It documents what might fail, why it matters, what controls exist and who will reduce the risk.

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TOPIC 03

When FMEA has the most leverage

1

New design or process

Prevent known patterns before approval or launch.

2

Changed use

Reassess risk when a product or process enters a new context.

3

Improvement goal

Anticipate failure introduced by the proposed change.

4

Before Control Plan

Convert prioritised risks into prevention and detection controls.

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TOPIC 03

Three common FMEA views

System FMEA

- Interfaces and interactions
- System functions
- Cross-boundary effects

Design FMEA

- Product functions
- Materials and geometry
- Design prevention

Process FMEA

- How production introduces defects
- 6M causes
- Feeds the Control Plan

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TOPIC 03

Prospective FMEA and retrospective RCA

Use

- Asks: what could fail?
- Acts before the next loss
- Uses functions, experience and change knowledge

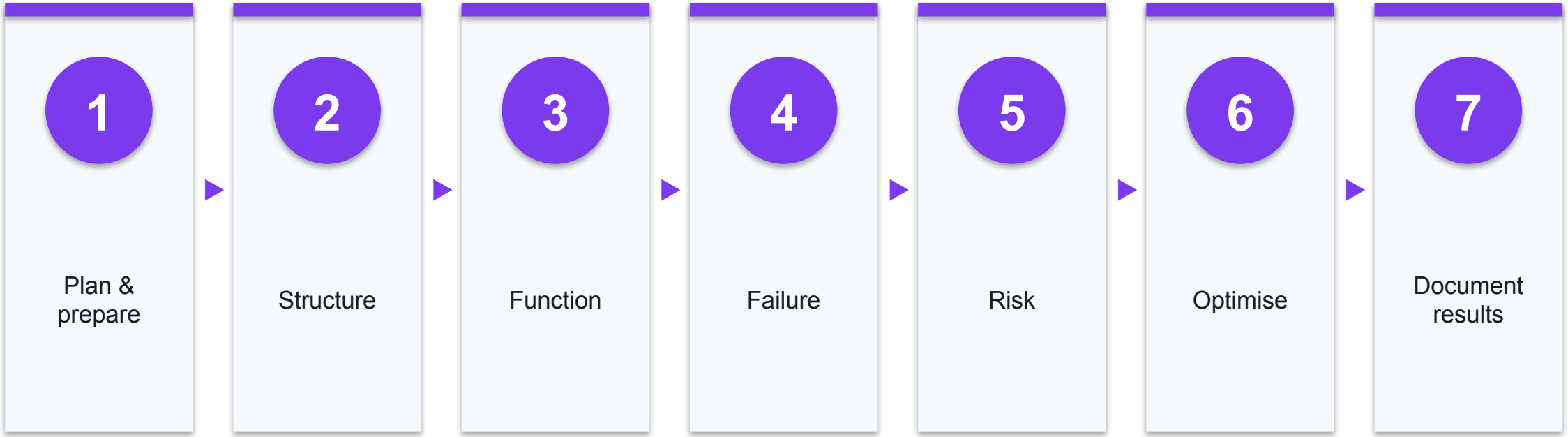
Contrast / check

- Asks: what happened and why?
- Acts after a failure signal
- Verified findings should update the FMEA

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TOPIC 03

Seven-step FMEA method



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Failure Analysis

METHOD STAGE 4 OF 7

4

Connect function, failure mode, effect and cause before assigning scores.

The exact workbook procedure is in Activity 05 and the Learner Guide.

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TOPIC 03

Step 1 — planning and preparation

1

Purpose

Name the decision the FMEA must support.

2

Scope

Set product/process boundaries and interfaces.

3

Team

Include design, process, quality, maintenance and operators.

4

Inputs

Gather flow maps, complaints, controls, changes and lessons.

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TOPIC 03

Steps 2–3 — structure and function

1

Structure

Break the process into logical steps and focus elements.

2

Interfaces

Show upstream, downstream and support interactions.

3

Function

Use verb–noun language with a measurable requirement.

4

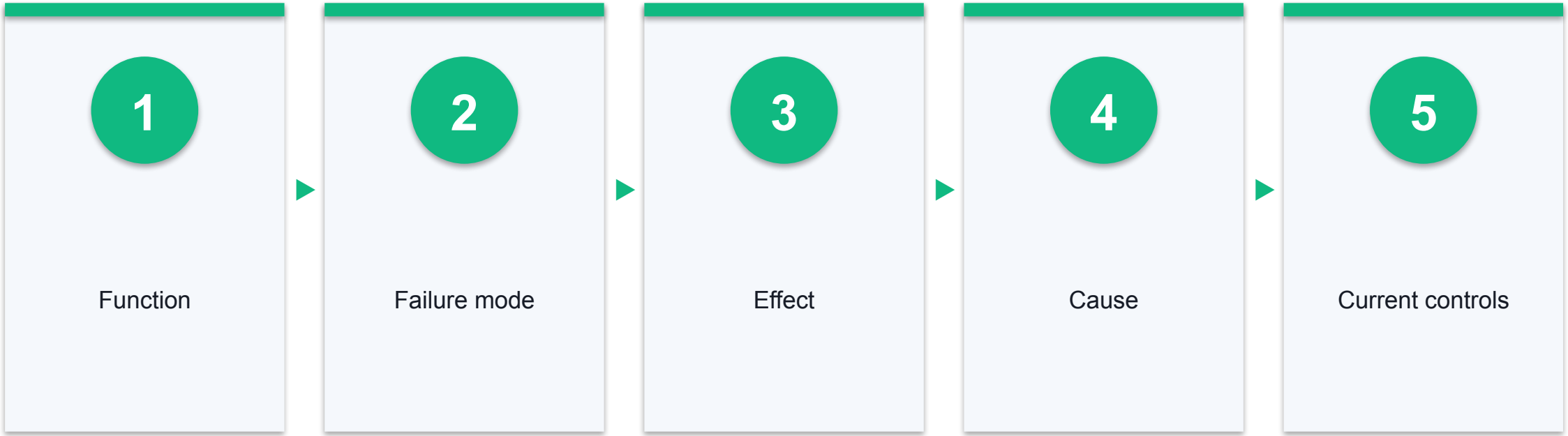
Requirement

State the expected output, tolerance or customer need.

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TOPIC 03

Step 4 — build the failure chain



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TOPIC 03

Write useful failure statements

1

Failure mode

Loss, partial, intermittent, excessive or unintended function.

2

Effect

What the next operation, system or customer experiences.

3

Cause

A specific mechanism or process condition that creates the mode.

4

Control

What prevents the cause or detects cause/mode before escape.

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TOPIC 03

Prevention and detection controls

Use

- Remove or reduce the cause
- Standard work, robust design, interlock
- Influences occurrence

Contrast / check

- Reveal cause or failure before escape
- Sensor, test, inspection, alarm
- Influences detection ranking

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TOPIC 03

Step 5 — S, O and D describe different risk dimensions

SEVERITY

- Consequence of the effect
- Score the most serious credible effect

OCCURRENCE

- Likelihood of the stated cause
- Use prevention and evidence

DETECTION

- Ability to detect before impact
- Higher score means weaker detection

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TOPIC 03

Severity rating discipline

1

Customer effect

Rate the most serious credible effect for the failure mode.

2

Consistent scale

Use one agreed 1–10 table and record the rationale.

3

Safety/regulatory

Escalate high-severity concerns regardless of a modest RPN.

4

Design change

Detection alone does not reduce the consequence severity.

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TOPIC 03

Occurrence rating discipline

1

Cause-specific

Rate the likelihood of the stated causal mechanism.

2

Use evidence

Prefer capability, defect, maintenance or comparable-history data.

3

Prevention matters

Strong prevention can reduce occurrence.

4

Assumptions visible

Record uncertainty and the basis for the score.

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TOPIC 03

Detection rating discipline

1

Higher means weaker

A high D score means the failure is hard to detect.

2

Control-specific

Rate the actual detection control, not intended inspection.

3

Before effect

Ask whether the cause or mode is caught before customer impact.

4

Evidence

Use validation, gauge reliability and escape history.

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TOPIC 03

RPN is useful—but incomplete

Calculate

- RPN = S × O × D
- Range 1–1000
- Useful for sorting and trending

Interpret

- Ordinal ratings
- Same RPN can hide different profiles
- Record scoring rationale

Guardrail

- No fixed cutoff as permission
- Severity first
- Use team judgement and Action Priority

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TOPIC 03

Same number, different risk

Use

- S=10, O=2, D=2
- RPN = 40
- Potentially safety-critical effect

Contrast / check

- S=5, O=4, D=2
- RPN = 40
- Different consequence and action logic

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TOPIC 03

Step 6 — optimise in the right order

1. Eliminate or reduce the effect
2. Prevent the cause / reduce occurrence
3. Improve detection
4. Assign owner and due date
5. Verify then re-rate

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TOPIC 03

Strong recommended actions

1

Specific

Names the change, location and failure mechanism.

2

Owned

One accountable owner and a realistic due date.

3

Verifiable

Defines evidence that proves implementation and effectiveness.

4

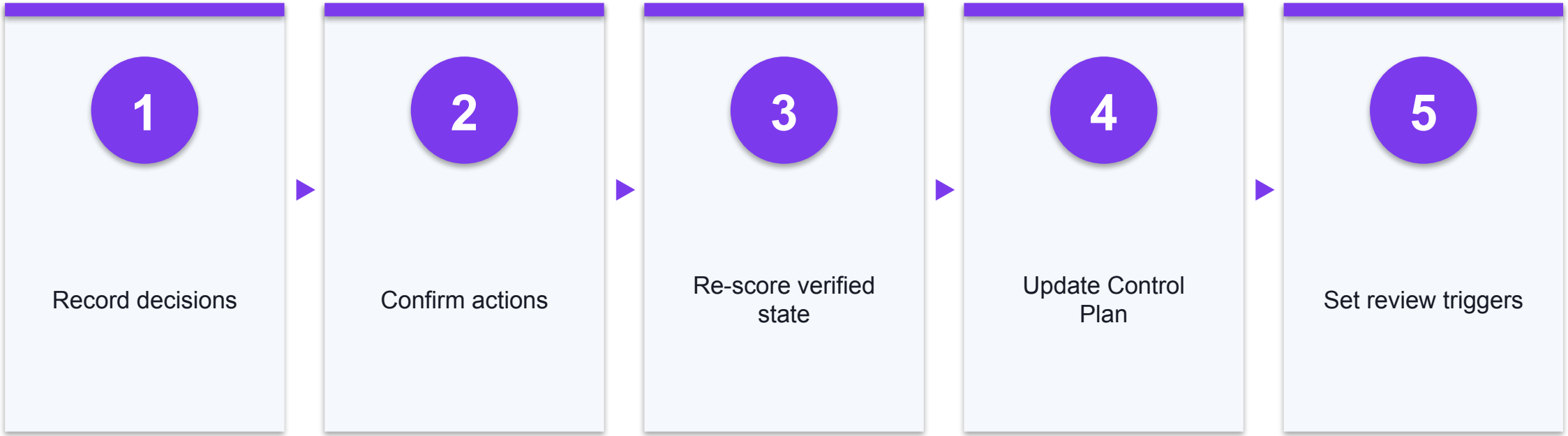
Traceable

Links to the PFMEA row, change record and Control Plan.

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TOPIC 03

Step 7 — document and sustain



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TOPIC 03

Review triggers for a living FMEA

1

Engineering change

New material, equipment, tooling, software or parameter.

2

Process signal

Capability loss, special cause, defect trend or downtime.

3

Customer signal

Complaint, return, field failure or changed requirement.

4

Learning

RCA, audit, maintenance finding or supplier feedback.

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TOPIC 03

FMEA pitfalls

1. **Scope creep**
   A boundary too broad prevents meaningful depth.
2. **Template completion**
   Scoring without insight or action creates false assurance.
3. **RPN gaming**
   Adjusting ratings to cross a threshold hides risk.
4. **Missing operators**
   The team loses actual-work failure knowledge.
5. **No ownership**
   Actions without owner/date remain recommendations.
6. **Stale document**
   A frozen FMEA stops reflecting the real process.

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ACTIVITY 05 · A5, A6, K3

Build the Orion Process FMEA

ORION SCENARIO

The team must turn the SIPOC, process flow, defect evidence and RCA findings into a traceable PFMEA.

YOUR MISSION

Create PFMEA rows from process function through effect, mode, cause and current prevention/detection controls.

DELIVERABLE

PFMEA header, process/function structure and at least eight complete failure chains

TOOLS AND FILES

PFMEA Excel template · S/O/D rating guides · evidence references

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Activity 05 — Decision Target

1

Objective

Apply structure, function and failure analysis to the scoped SMT process.

2

Question 1

Does each failure mode describe loss of a stated function?

3

Question 2

Which cause-control pair lacks evidence?

4

Evidence standard

Every row forms a coherent function–mode–effect–cause chain, controls are correctly classified, and evidence references are traceable.

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REVIEW BEFORE SUBMISSION

Activity 05 — Learning Check

✓ Acceptance check

Every row forms a coherent function–mode–effect–cause chain, controls are correctly classified, and evidence references are traceable.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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ACTIVITY 06 · A5, A6, K3

Rate Risk, Plan Actions and Re-score

ORION SCENARIO

Several Orion rows have similar RPNs but different customer and safety consequences.

YOUR MISSION

Rate S/O/D consistently, avoid threshold gaming, assign actions and verify risk reduction.

DELIVERABLE

Rated PFMEA with RPN formulas, internal action-priority guardrail, owners, due dates and post-action scores

TOOLS AND FILES

Excel formulas · conditional formatting · action register · verification evidence

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Activity 06 — Decision Target

1

Objective

Prioritise PFMEA actions using severity-first judgement, RPN context and verified re-rating.

2

Question 1

Why can an RPN of 40 require more urgent action than an RPN of 108?

3

Question 2

Which rating should change after adding only a better inspection?

4

Evidence standard

All RPN formulas are correct, severe risks remain visible, actions have ownership/evidence, and post-action ratings match the control change.

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REVIEW BEFORE SUBMISSION

Activity 06 — Learning Check

✓ Acceptance check

All RPN formulas are correct, severe risks remain visible, actions have ownership/evidence, and post-action ratings match the control change.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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TOPIC RECAP

Recap — Design and Apply FMEA

1

Concept

FMEA asks what could fail before the next customer or process loss occurs.

2

Decision

Function → failure mode → effect → cause → controls makes risk traceable.

3

Application

PFMEA header, process/function structure and at least eight complete failure chains

4

Transfer

Name one workplace risk, evidence source and next action.

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04

TOPIC 04

Identify and Direct Improvement Opportunities

Prioritise · select · pilot · coach · implement

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ELO4 · A7 · A8 · K2

Key Concepts — Improvement Opportunities

1

Target the failure mechanism

Prevention is stronger than inspection when the cause can be removed or constrained.

2

Choose transparently

Decision criteria make trade-offs visible and reduce preference-driven selection.

3

Pilot before scale

A bounded test checks benefit, side effects, capability and adoption.

4

Direct with ownership

Every action needs an owner, due date, success measure and escalation path.

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TOPIC 04 · WHY IT MATTERS

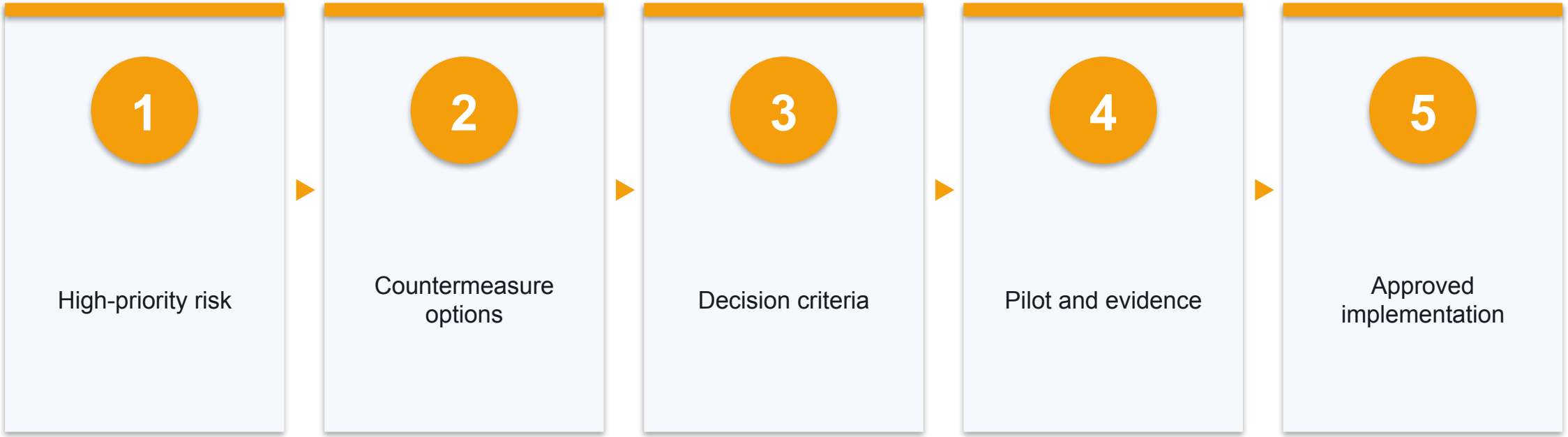
The best action changes the failure mechanism

Inspection may catch a defect; prevention changes why it occurs.

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TOPIC 04

From risk to improvement portfolio



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TOPIC 04

Countermeasure hierarchy

Prevent

- Eliminate the cause
- Robust design
- Poka-yoke or interlock

Reduce

- Improve capability
- Control variation
- Standard work and maintenance

Detect

- Automated checks
- Validated inspection
- Fast containment signal

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TOPIC 04

Opportunity sources

1

PFMEA

High severity, Action Priority, occurrence and weak detection.

2

Pareto and RCA

Recurring defects and verified mechanisms.

3

Capability and SPC

Unstable or incapable critical characteristics.

4

People and customers

Operator constraints, complaints and near misses.

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TOPIC 04

DMAIC provides a project roadmap

1

Define

Clarify the customer, problem, scope, target and ownership.

2

Measure

Validate definitions and establish a trustworthy baseline.

3

Analyse

Prioritise patterns and verify causal mechanisms.

4

Improve

Select, pilot and implement countermeasures.

5

Control

Monitor, react, standardise and transfer ownership.

6

Relationship to PDCA

Both are evidence-led cycles; DMAIC gives a detailed improvement-project structure.

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TOPIC 04

5S supports stable process conditions

1. **Sort**
   Remove items that are not needed for the work.
2. **Set in order**
   Make the correct item and sequence obvious.
3. **Shine**
   Clean while inspecting for early abnormality.
4. **Standardise**
   Define a repeatable condition and visual method.
5. **Sustain**
   Audit, coach and correct drift as normal work.

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TOPIC 04

DOE tests factors and interactions

Use

- Plan combinations of factor levels
- Estimate main effects and interactions
- Use randomisation, replication and blocking where appropriate

Contrast / check

- Do not change many factors informally
- Confirm measurement and process stability first
- Use statistical expertise for safety-critical or complex designs

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TOPIC 04

Evaluation criteria

1. **Risk reduction**
   Expected change to severity, occurrence or detection.
2. **Effectiveness**
   Strength of evidence and fit to the causal mechanism.
3. **Feasibility**
   Skills, time, technology and operational constraints.
4. **Cost and speed**
   Investment, lifecycle cost and time to benefit.
5. **Side effects**
   New failure modes, workload or compliance concerns.
6. **Sustainability**
   Ability to standardise, train, monitor and maintain.

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TOPIC 04

Weighted criteria matrix and Pugh matrix

Use

- Scores options against weighted criteria
- Useful when priorities differ in importance
- Shows a total and sensitivity to weights

Contrast / check

- Compares each option with a datum
- Uses better / same / worse reasoning
- Useful for concept screening and discussion

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TOPIC 04

Cost-benefit without false precision

1

Benefit

Avoided scrap, rework, downtime, complaints and risk exposure.

2

Cost

Capital, validation, training, maintenance and transition loss.

3

Time

Separate one-off savings from recurring annual impact.

4

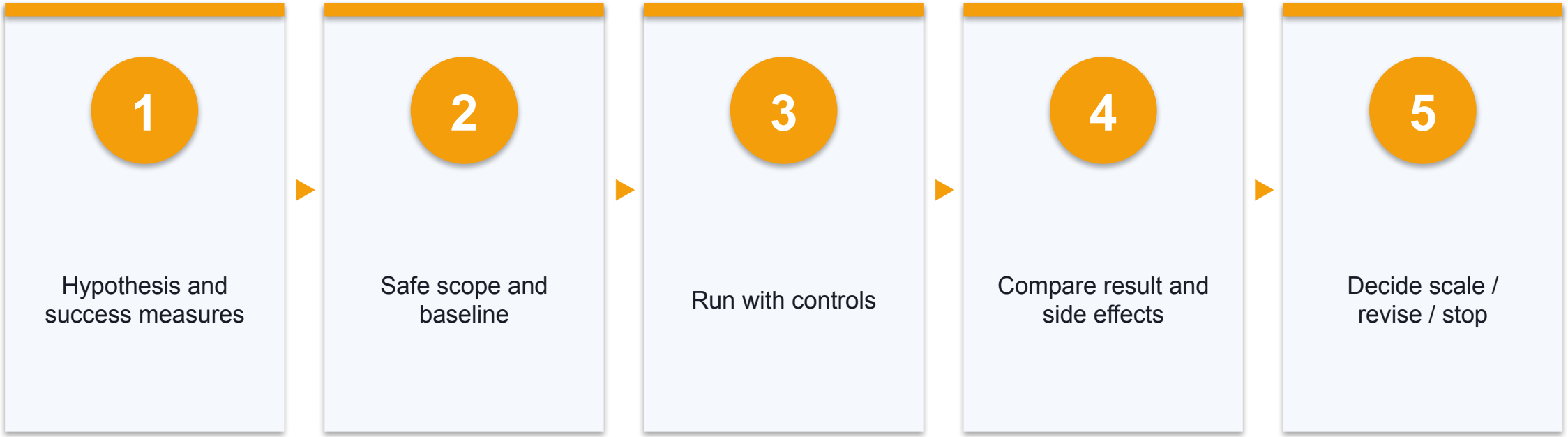
Uncertainty

Show assumptions and ranges instead of one confident number.

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TOPIC 04

Pilot learning cycle



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TOPIC 04

A credible pilot

1

Representative

Covers the relevant product, shift and failure conditions.

2

Contained

Limits customer and production risk during the test.

3

Measured

Uses defined input, process and outcome measures.

4

Reversible

Includes a stop rule and recovery plan.

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TOPIC 04

Poka-yoke strategies

Contact

- Physical fit or sensor
- Detects shape/position
- Prevents wrong assembly

Fixed value

- Counts required items or actions
- Signals omissions
- Useful for kits and fasteners

Motion sequence

- Enforces task order
- Interlock or checklist
- Prevents skipped steps

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TOPIC 04

Direct and coach the team

1

Purpose

Explain the customer/risk reason for the change.

2

Practice

Demonstrate, observe and let the operator perform.

3

Feedback

Correct the method and invite actual-work concerns.

4

Follow-up

Check adoption, outcome and unintended effects.

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TOPIC 04

Implementation control

1

Owner and date

One accountable person and milestone.

2

Change control

Approved specification, work instruction and training.

3

Verification

Evidence the action was installed as intended.

4

Effectiveness

Evidence it reduced risk without creating new failure.

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ACTIVITY 07 · A7, A8, K2

Select the Improvement Portfolio

ORION SCENARIO

Orion can fund only two of five proposed improvements during the next production quarter.

YOUR MISSION

Use weighted criteria, Pugh comparison and cost-benefit ranges to recommend a balanced action portfolio.

DELIVERABLE

Opportunity register, weighted decision matrix, Pugh matrix and cost-benefit summary

TOOLS AND FILES

Excel decision analysis · sensitivity check · PFMEA traceability

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Activity 07 — Decision Target

1

Objective

Identify and compare risk-reduction opportunities using transparent criteria.

2

Question 1

Does the top score remain top when cost weight doubles?

3

Question 2

Which option prevents the cause rather than merely detecting the defect?

4

Evidence standard

The recommendation traces to PFMEA risks, calculations are formula-driven, assumptions are visible and trade-offs are explicitly discussed.

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REVIEW BEFORE SUBMISSION

Activity 07 — Learning Check

✓ Acceptance check

The recommendation traces to PFMEA risks, calculations are formula-driven, assumptions are visible and trade-offs are explicitly discussed.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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ACTIVITY 08 · A7, A8, K2

Pilot, Direct and Coach the Change

ORION SCENARIO

Orion will pilot an automated stencil-cleaning trigger and first-piece verification on Printer P2.

YOUR MISSION

Design the pilot, success measures, reaction rules, standard-work change and operator coaching plan.

DELIVERABLE

Pilot charter, before/after measurement plan, risk guardrails, training and coaching checklist

TOOLS AND FILES

PDCA pilot · change control · MOC checklist · coaching observation

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Activity 08 — Decision Target

1

Objective

Plan a controlled pilot and direct team implementation with coaching and change safeguards.

2

Question 1

What would cause the pilot to stop immediately?

3

Question 2

How will the coach distinguish non-adoption from an ineffective countermeasure?

4

Evidence standard

The pilot is measurable, contained and reversible; the change has approvals, training and a documented scale/revise/stop rule.

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REVIEW BEFORE SUBMISSION

Activity 08 — Learning Check

✓ Acceptance check

The pilot is measurable, contained and reversible; the change has approvals, training and a documented scale/revise/stop rule.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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TOPIC RECAP

Recap — Identify and Direct Improvement Opportunities

1

Concept

Prevention is stronger than inspection when the cause can be removed or constrained.

2

Decision

Decision criteria make trade-offs visible and reduce preference-driven selection.

3

Application

Opportunity register, weighted decision matrix, Pugh matrix and cost-benefit summary

4

Transfer

Name one workplace risk, evidence source and next action.

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05

TOPIC 05

Manage Improvement with a Process Control Plan

Variation · capability · SPC · reaction · review

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ELO5 · A9 · A10 · K4

Key Concepts — Process Control Plan

1

Control the critical few

Translate high-risk PFMEA causes and characteristics into practical controls.

2

Know the variation

Common-cause variation needs system improvement; special causes need investigation.

3

React by design

A Control Plan must state who acts, what is contained and how the process restarts.

4

Close the loop

Control results, incidents and changes feed back into the PFMEA and standard work.

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TOPIC 05 · WHY IT MATTERS

Control is a feedback system

A stable process detects meaningful change, contains risk and learns before the customer does.

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TOPIC 05

Common-cause and special-cause variation

Use

- Built into the current system
- Many small sources
- Improve the process as a system

Contrast / check

- Specific, unusual signal
- Assignable event or condition
- Investigate and remove the cause

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TOPIC 05

Stability and capability answer different questions

1

Stability

Is the process behaviour predictable over time?

2

Capability

Can a stable process meet specification consistently?

3

Sequence

Check stability before interpreting capability indices.

4

Customer

Specifications come from requirements; control limits come from process data.

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TOPIC 05

Cp and Cpk

Cp

- Potential capability
- Specification width ÷ 6σ
- Assumes centring

Cpk

- Actual centring-aware capability
- Minimum distance to limit ÷ 3σ
- Cpk ≤ Cp

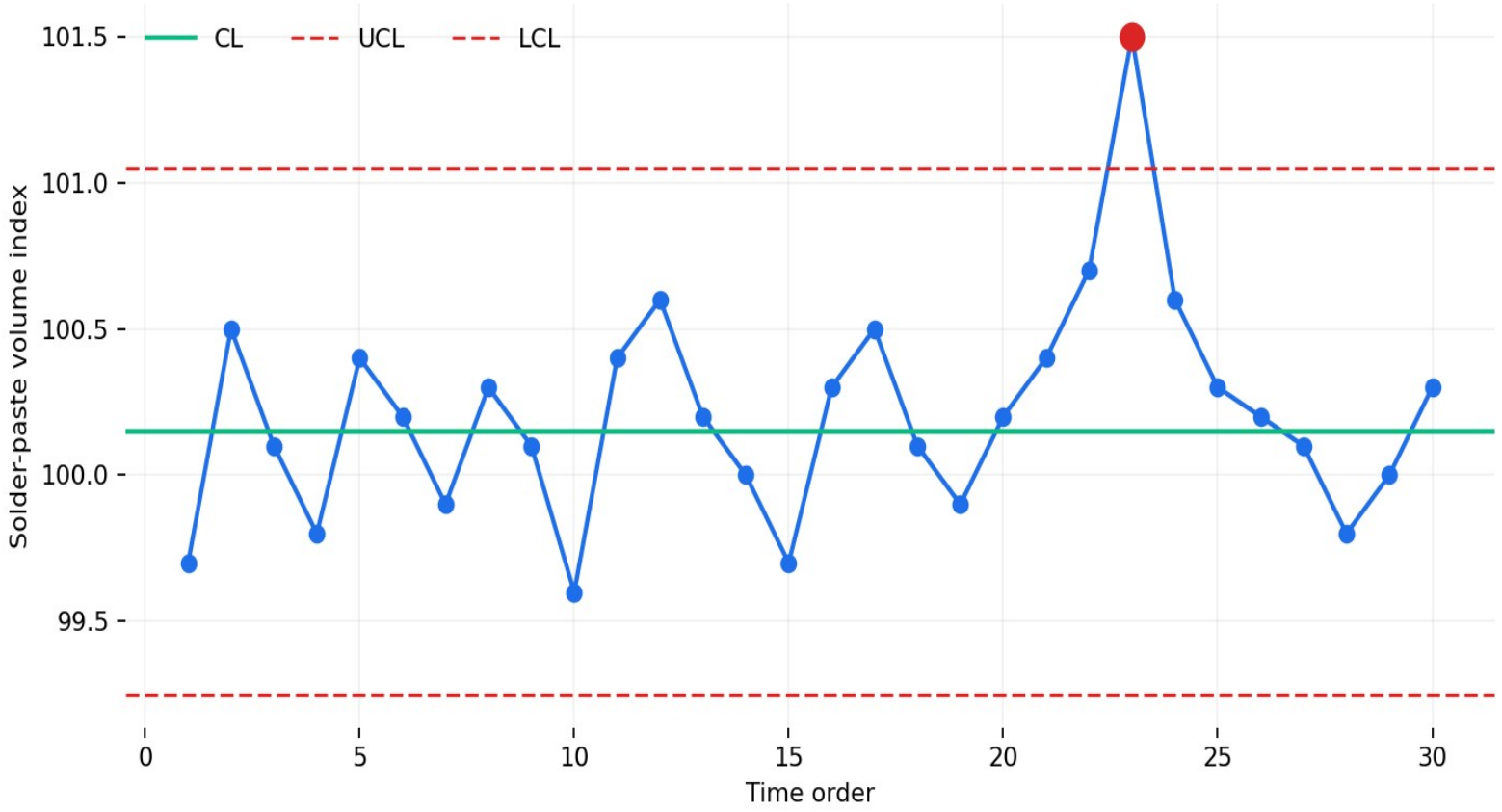
Interpret

- Use stable data
- Confirm distribution and measurement
- Report assumptions

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TOPIC 05

A control chart separates signal from noise



CONTROL ≠ SPECIFICATION

- Limits describe process behaviour.
- A signal prompts investigation.
- Capability compares stable variation with specifications.

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TOPIC 05

Control limits and specification limits

Use

- Calculated from process behaviour
- Support stability decisions
- Not changed to make the chart look better

Contrast / check

- Defined by customer/design requirement
- Support conformance decisions
- Do not describe natural process variation

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TOPIC 05

Control-chart signals

1

Point beyond a limit

Immediate special-cause investigation.

2

Run on one side

Possible process shift.

3

Trend

Gradual change, wear or drift.

4

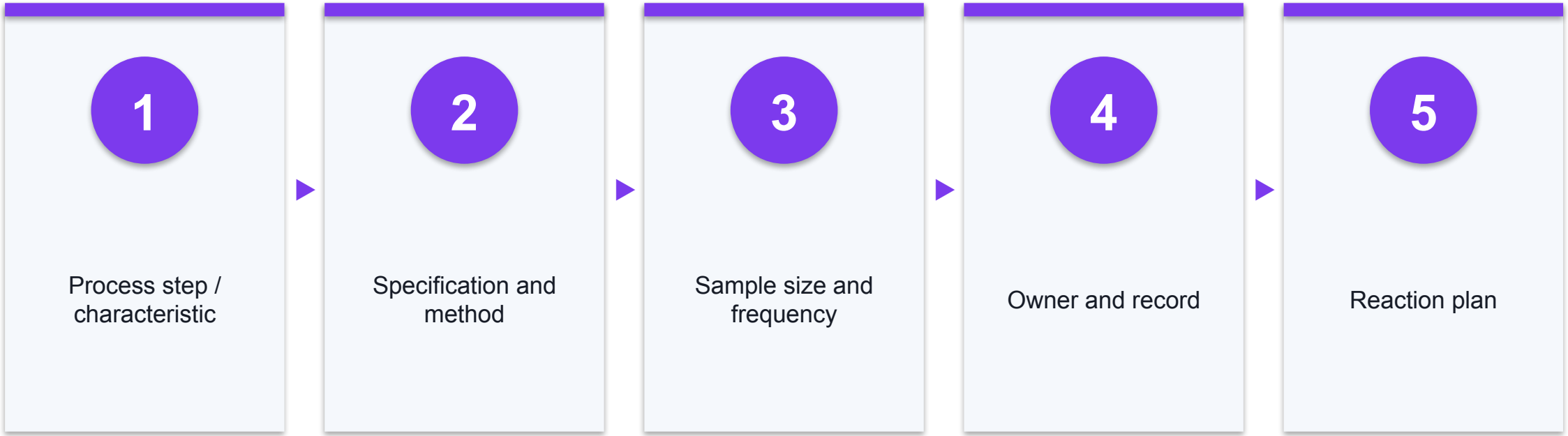
Cycle or clustering

Time pattern, batching or stratification.

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TOPIC 05

The Control Plan operationalises the PFMEA



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TOPIC 05

Control selection principles

1

Risk-based

Strength and frequency reflect severity, occurrence and escape risk.

2

At source

Control as close as possible to the causal process condition.

3

Capable measurement

The method can reliably distinguish acceptable from unacceptable.

4

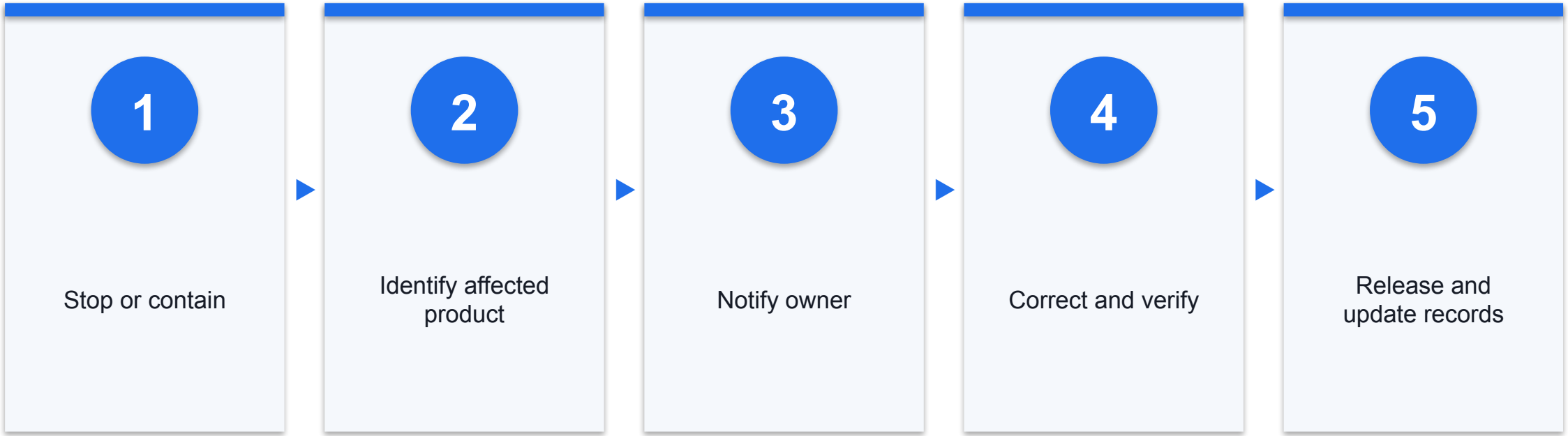
Actionable

A signal must trigger a defined response, not only a record.

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TOPIC 05

Reaction plan



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TOPIC 05

Monitoring and evaluation

Use

- Leading: parameter adherence, preventive maintenance, alarm rate
- Detect drift before outcome loss
- Useful for early intervention

Contrast / check

- Lagging: defect, scrap, rework, return, downtime
- Confirms realised outcome
- Useful for accountability and learning

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TOPIC 05

Evaluate an improvement

1. **Effectiveness**
   Did the targeted measure improve against baseline?

2. **Sustainment**
   Did the result persist across time, shift and product mix?

3. **Side effects**
   Did another defect, delay or workload appear?

4. **Adoption**
   Is standard work followed and understood?

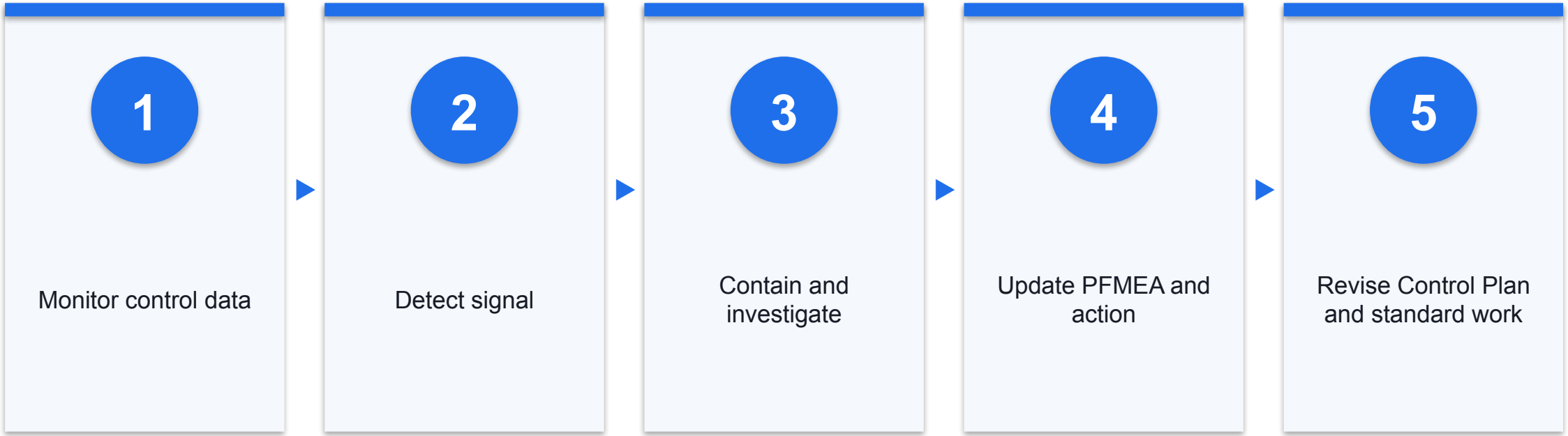
5. **Risk**
   Are PFMEA ratings and controls updated with evidence?

6. **Learning**
   What should be standardised, transferred or investigated next?

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TOPIC 05

Closed-loop improvement



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TOPIC 05

Governance review questions

1

Right risk?

Are high-severity and high-priority items visible?

2

Right evidence?

Are ratings, capability and results based on trusted data?

3

Right action?

Does the countermeasure address the verified mechanism?

4

Right control?

Will drift be detected and contained in time?

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ACTIVITY 09 · A9, A10, K4

Evaluate Stability and Capability

ORION SCENARIO

After the pilot, Orion has 30 daily solder-paste-volume samples and bridge-defect results.

YOUR MISSION

Build a control chart, investigate signals and calculate Cp/Cpk only after checking stability and assumptions.

DELIVERABLE

Time-series dataset, control chart, signal log, capability calculation and evaluation statement

TOOLS AND FILES

Excel AVERAGE · STDEV.S · UCL/LCL formulas · Cp/Cpk · trend interpretation

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Activity 09 — Decision Target

1

Objective

Monitor improvement performance using time order, control limits and capability measures.

2

Question 1

Why can a stable process still be incapable?

3

Question 2

What does Cpk below Cp reveal?

4

Evidence standard

Limits are formula-driven, signals are not confused with specification failures, capability is interpreted with stability and measurement caveats.

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REVIEW BEFORE SUBMISSION

Activity 09 — Learning Check

✓ Acceptance check

Limits are formula-driven, signals are not confused with specification failures, capability is interpreted with stability and measurement caveats.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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ACTIVITY 10 · A9, A10, K4

Create the Process Control Plan and Review Loop

ORION SCENARIO

Orion must hand the improved SMT process back to production without losing the risk knowledge created by the team.

YOUR MISSION

Create a deployable Control Plan linked to PFMEA rows, validated controls and governance review.

DELIVERABLE

Process Control Plan, reaction plan, audit checklist, control-to-PFMEA trace matrix and review calendar

TOOLS AND FILES

Excel Control Plan · PFMEA traceability · reaction plan · sustainment audit

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Activity 10 — Decision Target

1

Objective

Translate PFMEA risks into operational controls, reaction plans and FMEA review triggers.

2

Question 1

Which PFMEA risk has no adequate operational control?

3

Question 2

What event triggers immediate PFMEA review?

4

Evidence standard

Every critical PFMEA risk has an owned control and reaction plan, traceability is complete, and review triggers close the learning loop.

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REVIEW BEFORE SUBMISSION

Activity 10 — Learning Check

✓ Acceptance check

Every critical PFMEA risk has an owned control and reaction plan, traceability is complete, and review triggers close the learning loop.

PASS = complete · traceable · evidence-based · saved

EVIDENCE

- Completed workbook
- Recorded decision and rationale

SELF-CHECK

- Formula outputs reconcile
- Assumptions are visible
- Next action has an owner

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TOPIC RECAP

Recap — Manage Improvement with a Process Control Plan

1

Concept

Translate high-risk PFMEA causes and characteristics into practical controls.

2

Decision

Common-cause variation needs system improvement; special causes need investigation.

3

Application

Time-series dataset, control chart, signal log, capability calculation and evaluation statement

4

Transfer

Name one workplace risk, evidence source and next action.

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WRAP-UP

Course Summary & Assessment

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LEARNING OUTCOMES

What You Achieved

1

ELO1

Create a continuous process improvement system.

2

ELO2

Perform root cause analysis for continuous process improvement.

3

ELO3

Design and apply Failure Mode and Effects Analysis (FMEA).

4

ELO4

Identify and direct improvement opportunities.

5

ELO5

Manage and evaluate improvement with a Process Control Plan.

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SOURCE-ALIGNED COURSEWARE

Key FMEA References

ASQ	FMEA purpose, timing and team method
Jama Software	Seven-step method, Action Priority and RPN cautions
Quality-One	Functions, controls, action ownership and re-rating
PMC	FMEA as a prospective safety and risk-control tool
IPH Hannover	German industrial FMEA practice and prevention context
ScienceDirect	Engineering overview and related reliability concepts

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FINAL REMINDER

Assessment Conditions

1

Individual

Complete both assessment instruments independently.

2

Approved resources

Use only the course materials allowed by the assessor.

3

Evidence

Submit the exact evidence named in each task.

4

Clarification

Ask about instructions—not answers.

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Assessment Flow

1. Assessment attendance
2. Written Assessment · K1–K4 · 60 min
3. Practical Performance · A1–A10 · 90 min
4. Submit candidate evidence
5. Sign Assessment Summary Record

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Digital Attendance (Mandatory)

1

Assessment attendance

Scan before the assessment starts.

2

Competency record

Complete WA and PP and sign the summary record.

3

Submission

Upload only the required learner evidence.

4

Support

Ask the assessor about re-assessment or appeal if required.

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FROM RISK TO CONTROL

Thank You!

Prevent failure, improve the system and make learning visible.

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